CSE 715: Computer Graphics

Chapter 6 — Three-Dimensional Transformations

Complete Model Answers (2020–2024)

University of Chittagong — 7th Semester B.Sc. (Engg.) Examination

Source: Schaum's Outline of Computer Graphics (Xiang & Plastock), Chapter 6 — notation follows the book exactly ($R_{\theta,\mathbf{I}}/R_{\theta,\mathbf{J}}/R_{\theta,\mathbf{K}}$ for rotation about $x/y/z$, S_{s_x,s_y,s_z} for scaling, $T_{\mathbf{V}}$ for translation)

Contents

Theoretical Framework — Reusable Across All Years

Yield note: Chapter 6 is the lightest exam chapter — across all 5 years (2020–2024), only **one** genuine Chapter-6 question appears: **tilting** (2022, Q5c, 3.5 marks). **3D scaling** (2023, Q3c, 4 marks) is Chapter-4 scaling machinery applied to 3D points and is already solved in Chapter4_Solutions.pdf — it is re-solved here too since it is Ch6 material by chapter number. **Mirror reflection** is syllabus-mandated ("Chapter 6: rotation, transformation, scaling, mirror") but has **0/5 PYQ hits** — included below for completeness only, same treatment as Bezier curves in Ch9.

Translation, Scaling, and Canonical-Axis Rotation (§6.1)

Translation by $\mathbf{V} = a\mathbf{I} + b\mathbf{J} + c\mathbf{K}$: $x' = x + a,\ y' = y + b,\ z' = z + c$.

$$T_{\mathbf{V}} = \begin{pmatrix} 1 & 0 & 0 & a \\ 0 & 1 & 0 & b \\ 0 & 0 & 1 & c \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Scaling about the origin with factors s_x, s_y, s_z ($s > 1$ magnifies, $s < 1$ reduces): $x' = s_x \cdot x,\ y' = s_y \cdot y,\ z' = s_z \cdot z$.

$$S_{s_x,s_y,s_z} = \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{pmatrix}$$

Canonical rotations (right-hand rule, positive angle θ about the named axis):

$$R_{\theta,\mathbf{K}} \text{ (about } z) : \begin{cases} x' = x\cos\theta - y\sin\theta \\ y' = x\sin\theta + y\cos\theta \\ z' = z \end{cases} \qquad R_{\theta,\mathbf{J}} \text{ (about } y) : \begin{cases} x' = x\cos\theta + z\sin\theta \\ y' = y \\ z' = -x\sin\theta + z\cos\theta \end{cases} \qquad R_{\theta,\mathbf{I}} \text{ (ab}$$

$$R_{\theta,\mathbf{K}} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad R_{\theta,\mathbf{J}} = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \quad R_{\theta,\mathbf{I}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

Coordinate transformations (move the observer, object fixed) use the same matrices with $\theta \to -\theta$ and $\mathbf{V} \to -\mathbf{V}$ (e.g. $\bar{T}_{\mathbf{V}} = T_{-\mathbf{V}}$, $\bar{R}_\theta = R_{-\theta}$) — direct 3D analogue of Ch4's geometric-vs-coordinate distinction. **Homogeneous form:** all 3×3 matrices above adjoin a 4th row/column $(0\,0\,0\,1)$ so rotation, scaling, and translation can be concatenated by matrix multiplication (order matters — rightmost matrix applies first to a column vector).

Composite / Tilting Transformation (§6.3, Solved Problem 6.1)

Tilting is defined as a rotation about the x-axis (angle θ_x) followed by a rotation about the y-axis (angle θ_y). Since the x-rotation is applied first, it sits closest to the vector being transformed:

$$T = R_{\theta_y,\mathbf{J}} \cdot R_{\theta_x,\mathbf{I}} = \begin{pmatrix} \cos\theta_y & \sin\theta_y\sin\theta_x & \sin\theta_y\cos\theta_x \\ 0 & \cos\theta_x & -\sin\theta_x \\ -\sin\theta_y & \cos\theta_y\sin\theta_x & \cos\theta_y\cos\theta_x \end{pmatrix}$$

Does order matter? Reversing the order (rotate about y first, then x): $R_{\theta_x,\mathbf{I}} \cdot R_{\theta_y,\mathbf{J}} = \begin{pmatrix} \cos\theta_y & 0 & \sin\theta_y \\ \sin\theta_x\sin\theta_y & \cos\theta_x & -\sin\theta_x\cos\theta_y \\ -\cos\theta_x\sin\theta_y & \sin\theta_x & \cos\theta_x\cos\theta_y \end{pmatrix}$ — **not the same matrix as** T (compare position (2,1): 0 vs. $\sin\theta_x\sin\theta_y$). **Matrix multiplication does not commute in general ⇒ order matters** for 3D composite rotations, unlike scalar multiplication.

Rotation About an Arbitrary Axis — Alignment (§6.3, Solved Problems 6.2–6.4)

To rotate by θ about an arbitrary axis L (direction $\mathbf{V} = a\mathbf{I} + b\mathbf{J} + c\mathbf{K}$, through point P): (1) translate P to the origin (T_{-P}); (2) align $\mathbf{V}$ with $\mathbf{K}$ via $A_{\mathbf{V}}$ (two canonical rotations: about x by θ_1 with $\sin\theta_1 = \frac{b}{\sqrt{b^2+c^2}}, \cos\theta_1 = \frac{c}{\sqrt{b^2+c^2}}$, then about y by $-\theta_2$ with $\lambda = \sqrt{b^2+c^2}$, $\sin(-\theta_2) = \frac{-a}{|\mathbf{V}|}, \cos(-\theta_2) = \frac{\lambda}{|\mathbf{V}|}$); (3) rotate by θ about $\mathbf{K}$ ($R_{\theta,\mathbf{K}}$); (4) reverse steps 2 and 1:

$$R_{\theta,L} = T_{-P}^{-1} \cdot A_{\mathbf{V}}^{-1} \cdot R_{\theta,\mathbf{K}} \cdot A_{\mathbf{V}} \cdot T_{-P}$$

Mirror Reflection (§6.3, Solved Problems 6.6–6.8) — syllabus-mandated, 0/5 PYQ

Reflection about the xy-plane: $P(x, y, z) \to P'(x, y, -z)$, i.e. $M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$.

Reflection about an arbitrary plane through $P_0(x_0, y_0, z_0)$ with normal $\mathbf{N}$: translate P_0 to the origin, align $\mathbf{N}$ with $\mathbf{K}$, apply the xy-plane reflection M, then reverse: $M_{\mathbf{N},P_0} = T_{\mathbf{V}}^{-1} \cdot A_{\mathbf{N}}^{-1} \cdot M \cdot A_{\mathbf{N}} \cdot T_{\mathbf{V}}$ where $\mathbf{V} = -x_0\mathbf{I} - y_0\mathbf{J} - z_0\mathbf{K}$.

Examination 2022 — Q5(c) (Section B): Tilting

Question: Define tilting as a rotation about the x-axis followed by a rotation about the y-axis. (i) Find the tilting matrix. (ii) Does the order of performing the rotation matter? [3.5 marks]

Solution

Definition: tilting = rotate about the x-axis by θ_x, then rotate the result about the y-axis by θ_y.

(i) Tilting matrix — x-rotation applied first, so it is rightmost in the product:

$$T = R_{\theta_y,\mathbf{J}} \cdot R_{\theta_x,\mathbf{I}} = \begin{pmatrix} \cos\theta_y & 0 & \sin\theta_y \\ 0 & 1 & 0 \\ -\sin\theta_y & 0 & \cos\theta_y \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_x & -\sin\theta_x \\ 0 & \sin\theta_x & \cos\theta_x \end{pmatrix} = \begin{pmatrix} \cos\theta_y & \sin\theta_y \sin\theta_x & \sin\theta_y \cos\theta_x \\ 0 & \cos\theta_x & -\sin\theta_x \\ -\sin\theta_y & \cos\theta_y \sin\theta_x & \cos\theta_y \cos\theta_x \end{pmatrix}$$

(ii) Does order matter? Reverse the order — rotate about y first, then x:

$$R_{\theta_x,\mathbf{I}} \cdot R_{\theta_y,\mathbf{J}} = \begin{pmatrix} \cos\theta_y & 0 & \sin\theta_y \\ \sin\theta_x \sin\theta_y & \cos\theta_x & -\sin\theta_x \cos\theta_y \\ -\cos\theta_x \sin\theta_y & \sin\theta_x & \cos\theta_x \cos\theta_y \end{pmatrix}$$

This differs from T (e.g. entry $(2,1)$ is 0 in T but $\sin\theta_x \sin\theta_y$ here). **Yes, the order matters** — matrix multiplication is not commutative, so ”rotate about x then y” and ”rotate about y then x” give physically different final orientations.

Examination 2023 — Q3(c) (Section A): 3D Scaling (Related — also in Chapter 4)

Question: Given a 3D object with coordinate points $A(3,0,3)$, $B(3,3,6)$, $C(3,0,1)$, and $D(0,0,0)$. Apply the scaling parameter 3 towards the X axis, 2 towards the Y axis, and 3 towards the Z axis, and obtain the new coordinates of the object. [4 marks]

Solution

Scaling about the origin with $s_x = 3,\ s_y = 2,\ s_z = 3$: $x' = 3x,\ y' = 2y,\ z' = 3z$.

$$S_{3,2,3} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

Vertex	Original (x, y, z)	Apply $(3x, 2y, 3z)$	New (x', y', z')
A	$(3, 0, 3)$	$(9, 0, 9)$	$\mathbf{A'(9, 0, 9)}$
B	$(3, 3, 6)$	$(9, 6, 18)$	$\mathbf{B'(9, 6, 18)}$
C	$(3, 0, 1)$	$(9, 0, 3)$	$\mathbf{C'(9, 0, 3)}$
D	$(0, 0, 0)$	$(0, 0, 0)$	$\mathbf{D'(0, 0, 0)}$

Identical result to Chapter4_Solutions.pdf's treatment of this same question (2D-scaling machinery generalizes directly — each axis just scales independently).

Quick Summary — Exam Patterns for Chapter 6

Pattern	**Years**	**Method to memorise**
Tilting (composite rotation: x then y) + does order matter	2022 only	$T = R_{\theta_y,\mathbf{J}} \cdot R_{\theta_x,\mathbf{I}}$; multiply the two canonical matrices, plug in, then show the reversed product differs $\Rightarrow$ order matters
3D scaling application about the origin	2023 only	Multiply each coordinate independently by its axis scale factor s_x, s_y, s_z — identical to 2D scaling, one more axis
Mirror reflection (syllabus-mandated, never asked)	0/5	xy-plane: negate z. Arbitrary plane: translate-align-reflect-reverse, same recipe as arbitrary-axis rotation

Exam-day strategy for Chapter 6: correction from earlier tracking — tilting was previously logged as a 2/5-year topic (2021+2022); re-verified directly against all 5 raw scanned papers on 2026-07-05 and it only appears in **2022 (Q5c)**. Given the entire chapter is worth at most ∼4 marks in a typical year and sometimes 0, this is a "learn the method once, move on" chapter — don't over-invest revision time here relative to Ch5/Ch7/Ch10.